www.tntextbooks.in

deserts. There are a variety of species that have been adapted successfully to live in diverse ecosystems. Ecosystem is a community of living organisms (plants and animals), non-living environment (including minerals, climate, soil, water, sunlight) and their interrelationships (**A.G. Tansley, 1935**). The presence of a large number of species in a particular ecosystem is called 'biological diversity' or in short **'biodiversity'.** The term biodiversity was first introduced by **Walter Rosen (1985),** and defined by E.D. Wilson.

Difference between the Living and Non-living

Living organisms show a variety of unique characters different from non-living matter. The key characters of living organisms are, cellular organization, nutrition, respiration, metabolism, growth, response to stimuli, movement, reproduction, excretion, adaptation and homeostasis. Numerous scientists and taxonomists have made tremendous contribution and documentation in the observation and study of even minute characters in living organisms. Their keen observations have led to the classification of living organisms and the study of their interrelationships.

1.2. Need for classification

We come across many places where things are arranged in specific categories. In super markets, the shelves can have rows and columns of groceries, cosmetics, toys, stationeries, snacks and utensils. If it is not arranged in a well organized manner, customers and sales persons will waste lot of time in finding an item. In the same way, libraries also organize the books alphabetically or genres-wise into autobiographies, novels, kids stories, science fictions, etc. Likewise it is nearly impossible to study all the living organism hence it becomes necessary to device some means and methods to make this possible and this process is called classification. Classification is a process by which things are grouped in convenient categories, based on easily observable characters. The scientific term used for these categories is **taxa** (taxon–singular). Taxa indicates categories at different levels, for example Kingdom Animalia, includes multicellular animals such as reptiles, mammals, etc. Based on their characteristics, all living organisms can be classified into different taxa. This science of classification is called **taxonomy**. External and internal structures along with developmental processes and ecological information of organisms are essential, as they form basis of the taxonomical studies. Hence, characterisation, identification, nomenclature and classification are the scientific stages that are basic to taxonomy.

The basic need for classifications are:

- To identify and differentiate closely related species
- To know the variation among the species
- To understand the evolution of the species
- To create a phylogenetic tree among the different groups
- To conveniently study living organisms